

Output Analysis - Statistical Methodology

1. Overview

This document specifies the statistical framework used to analyse results from the EMS Readiness Optimization production experiments (Phase 5). The analysis covers **2,760 simulation runs** across five experiment sets (Policy & Fleet Analysis, Demand Sensitivity, Service Robustness, CBD Robustness, and Capacity Sensitivity) and three allocation policies (P0 Spatially-Stratified Uniform, P1 Demand-Proportional, P2 Demand-Weighted Optimized). All procedures follow publication standards for discrete-event simulation output analysis (Law, 2015; Banks et al., 2010).

Cross-reference: See `docs/core/technical_report.md` §4.5.1 for the full factorial design table and `docs/core/experimental_design.md` for detailed factor definitions.

2. Primary Metrics

Metric	Symbol	Definition
Mean Response Time	$\bar{R}$	Average minutes from call arrival to unit arrival on scene
P90 (90th %ile) Response Time	R_{90}	90th percentile response time — time exceeded by only 10% of calls
6-min Coverage (NYC law)	C_6	Fraction of calls responded to within 6 minutes
8-min Coverage (NFPA standard)	C_8	Fraction of calls responded to within 8 minutes
Mean Utilization	$\bar{U}$	Average fraction of time units are busy
Mean Queue Length	$\bar{L}$	Time-weighted average queue length
Queue Fraction	Q_f	Fraction of incidents that waited in queue

CBD-Specific Metrics (CBD Robustness experiment only)

Metric	Symbol	Definition
CBD Mean RT	$\bar{R}_{\text{CBD}}$	Mean response time for CBD-zone incidents only
CBD 6-min Coverage	$C_{6,\text{CBD}}$	6-minute coverage within the CBD
CBD 8-min Coverage	$C_{8,\text{CBD}}$	8-minute coverage within the CBD

Capacity Sensitivity Metrics (Capacity experiment only)

Capacity sensitivity analysis (cap 1-5) uses the same primary metrics above, evaluated at $K \in \{20, 40\}$ with 15 replications per cell. See [docs/analysis/capacity_sensitivity_analysis.md](#).

3. Hypothesis Testing Framework

3.1 Research Hypotheses

ID	Hypothesis	Test
H1	Allocation policy significantly affects mean response time	One-way ANOVA
H2	Policy effect interacts with fleet size K	Two-way ANOVA (Policy $\times$ K)
H3	Policy effect interacts with demand level	Two-way ANOVA (Policy $\times$ Demand)
H4	Policy effect is robust to service-time variability	Two-way ANOVA (Policy $\times$ Service Time)

3.2 Significance Level

All tests use $\alpha = 0.05$ (two-sided). Family-wise error is controlled with Bonferroni or Tukey adjustments where multiple comparisons are made.

4. ANOVA Design and Assumptions

4.1 One-Way ANOVA (Experiment 1)

- **Factor:** Policy $\in \{P0, P1, P2\}$
- **Response:** Mean response time per replication
- **Replications:** 30 per policy (Common Random Numbers)

- **Assumptions checked:**
- Normality: Shapiro-Wilk test on residuals
- Homogeneity of variance: Levene’s test
- Independence: ensured by CRN design

4.2 Two-Way ANOVA (Experiments 2-4)

Experiment	Factor A	Factor B	Levels
Exp 2	Policy (P0, P1, P2)	K (15, 20, 25, 30, 35, 40)	3 × 6 = 18 cells
Exp 3	Policy (P0, P1, P2)	Demand (0.5, 0.75, 1.0, 1.25, 1.5, 2.0)	3 × 6 = 18 cells
Exp 4	Policy (P0, P1, P2)	Service Time (20, 25, 30)	3 × 3 = 9 cells

- **Model:** $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon$
- **Interaction term** $(\alpha\beta)_{ij}$ tests whether the policy advantage changes across factor levels.

4.3 Assumption Diagnostics

Assumption	Test	Action if Violated
Normality of residuals	Shapiro-Wilk (per group)	Use Kruskal-Wallis non-parametric alternative
Homogeneity of variance	Levene’s test	Use Welch’s ANOVA or report with caveat
Independence	Design-based (CRN pairs)	Paired analysis as sensitivity check

5. Multiple Comparison Procedures

5.1 Tukey’s Honest Significant Difference (HSD)

Used for **all pairwise** policy comparisons after a significant omnibus F-test. Controls family-wise Type I error at $\alpha = 0.05$ across all $\binom{k}{2}$ pairs.

$$q_{\text{HSD}} = q_{\alpha,k,N-k} \cdot \sqrt{\frac{MSE}{n}}$$

5.2 Bonferroni Correction

Applied when comparisons are pre-planned or when Tukey assumptions are questionable. Adjusted significance level: $\alpha' = \alpha / m$ where m is the number of comparisons.

6. Confidence Interval Methodology

6.1 Standard CIs

For each policy-scenario combination:

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

with $n = 30$ replications.

6.2 CIs for Differences

For paired policy comparisons (using CRN):

$$\bar{D} \pm t_{\alpha/2, n-1} \frac{s_D}{\sqrt{n}}$$

where $D_i = X_i^A - X_i^B$ is the paired difference for replication i .

6.3 Bootstrap CIs

If normality is rejected, we compute bias-corrected and accelerated (BCa) bootstrap intervals with 10 000 resamples.

7. Effect Size Interpretation

7.1 Cohen's d (Pairwise)

$$d = \frac{\bar{X}_A - \bar{X}_B}{s}$$

d	Interpretation
< 0.2	Negligible
0.2-0.5	Small
0.5-0.8	Medium
> 0.8	Large

7.2 Eta-squared (η^2) for ANOVA

$$\eta^2 = \frac{SS_{\text{effect}}}{SS_{\text{total}}}$$

η^2	Interpretation
< 0.01	Negligible
0.01-0.06	Small
0.06-0.14	Medium
> 0.14	Large

8. Publication Standards

- All p-values reported to 3 decimal places; $p < 0.001$ where applicable.
 - Effect sizes reported alongside p-values (APA 7th ed.).
 - Figures at 300 DPI, vector-quality where feasible.
 - Tables include 95% CIs in parentheses.
 - Multiple-comparison adjustments stated explicitly.
 - Statistical software: Python 3 with SciPy 1.x, statsmodels 0.14+.
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9. References

- Banks, J. et al. (2010). Discrete-Event System Simulation, 5th ed.
- Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences.
- Law, A. M. (2015). Simulation Modeling and Analysis, 5th ed.
- Tukey, J. W. (1977). Exploratory Data Analysis.